ECE 473 / 573: Cloud Computing and Cloud Native Systems

Homework 04: Task Scheduling for Resource Management

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Course: ECE 473 / 573: Cloud Computing and Cloud Native Systems (Fall 2026)
Assignment: Homework 04: Task Scheduling for Resource Management
Due Date: November 01, 2026 (Sunday, end of day Chicago time)

System Configuration and Task Arrival Profile

At time t = 0, five independent computational tasks arrive simultaneously in the ready queue. The expected execution burst times are:
• Task A: 6 seconds
• Task B: 8 seconds
• Task C: 4 seconds
• Task D: 10 seconds
• Task E: 5 seconds
Total CPU workload is 33 seconds. All tasks share an arrival time of t = 0 (A_i = 0 for all i).

Task Name	Arrival Time (t)	Expected Execution Time (Burst)	Initial Queue Position
Task A	0 seconds	6 seconds	Position 1
Task B	0 seconds	8 seconds	Position 2
Task C	0 seconds	4 seconds	Position 3
Task D	0 seconds	10 seconds	Position 4
Task E	0 seconds	5 seconds	Position 5

Question 1: Task Completion Times Across Scheduling Policies

For each scheduling policy, the step by step execution timeline and resulting task completion times are determined as follows:

1. First-Come, First-Served (FCFS)

FCFS schedules tasks strictly in order of arrival: A -> B -> C -> D -> E. Because all tasks arrive at t = 0, the CPU processes each task to completion in sequential arrival order without preemption:
• Task A executes from t = 0 to t = 6. Completion time C_A = 6 seconds.
• Task B executes from t = 6 to t = 14. Completion time C_B = 14 seconds.
• Task C executes from t = 14 to t = 18. Completion time C_C = 18 seconds.
• Task D executes from t = 18 to t = 28. Completion time C_D = 28 seconds.
• Task E executes from t = 28 to t = 33. Completion time C_E = 33 seconds.

2. Shortest Job First (SJF)

SJF schedules tasks in ascending order of expected execution time: C (4s) -> E (5s) -> A (6s) -> B (8s) -> D (10s). The scheduler prioritizes short jobs to clear the queue rapidly:
• Task C executes from t = 0 to t = 4. Completion time C_C = 4 seconds.
• Task E executes from t = 4 to t = 9. Completion time C_E = 9 seconds.

• Task A executes from t = 9 to t = 15. Completion time C_A = 15 seconds.
• Task B executes from t = 15 to t = 23. Completion time C_B = 23 seconds.
• Task D executes from t = 23 to t = 33. Completion time C_D = 33 seconds.

3. Longest Job First (LJF)

LJF schedules tasks in descending order of expected execution time: D (10s) -> B (8s) -> A (6s) -> E (5s) -> C (4s). Heavy compute jobs are processed first:
• Task D executes from t = 0 to t = 10. Completion time C_D = 10 seconds.
• Task B executes from t = 10 to t = 18. Completion time C_B = 18 seconds.
• Task A executes from t = 18 to t = 24. Completion time C_A = 24 seconds.
• Task E executes from t = 24 to t = 29. Completion time C_E = 29 seconds.
• Task C executes from t = 29 to t = 33. Completion time C_C = 33 seconds.

4. Round Robin (RR, Quantum q = 3 seconds)

Round Robin allocates a fixed 3-second time slice to each task following initial arrival order [A, B, C, D, E]. When a task completes its 3-second slice without finishing, the scheduler preempts it and appends it to the back of the ready queue:

Round 1:
• Task A runs from t = 0 to t = 3 (3s remaining). Queue: [B, C, D, E, A]
• Task B runs from t = 3 to t = 6 (5s remaining). Queue: [C, D, E, A, B]
• Task C runs from t = 6 to t = 9 (1s remaining). Queue: [D, E, A, B, C]
• Task D runs from t = 9 to t = 12 (7s remaining). Queue: [E, A, B, C, D]
• Task E runs from t = 12 to t = 15 (2s remaining). Queue: [A, B, C, D, E]

Round 2:
• Task A runs from t = 15 to t = 18 (0s remaining). Task A completes at t = 18 seconds. Queue: [B, C, D, E]
• Task B runs from t = 18 to t = 21 (2s remaining). Queue: [C, D, E, B]
• Task C runs from t = 21 to t = 22 (0s remaining, needs only 1s). Task C completes at t = 22 seconds. Queue: [D, E, B]
• Task D runs from t = 22 to t = 25 (4s remaining). Queue: [E, B, D]
• Task E runs from t = 25 to t = 27 (0s remaining, needs only 2s). Task E completes at t = 27 seconds. Queue: [B, D]

Round 3:
• Task B runs from t = 27 to t = 29 (0s remaining, needs only 2s). Task B completes at t = 29 seconds. Queue: [D]
• Task D runs from t = 29 to t = 32 (1s remaining). Queue: [D]

Round 4:
• Task D runs from t = 32 to t = 33 (0s remaining, needs only 1s). Task D completes at t = 33 seconds.

Task Name	Burst Time	FCFS Completion	SJF Completion	LJF Completion	Round Robin (q = 3)
Task A	6 seconds	6 seconds	15 seconds	24 seconds	18 seconds

Task B	8 seconds	14 seconds	23 seconds	18 seconds	29 seconds
Task C	4 seconds	18 seconds	4 seconds	33 seconds	22 seconds
Task D	10 seconds	28 seconds	33 seconds	10 seconds	33 seconds
Task E	5 seconds	33 seconds	9 seconds	29 seconds	27 seconds

Question 2: Average Turnaround Time Calculation and Policy Evaluation

Turnaround time (T_i) is defined as the elapsed time from task arrival to task completion: $T_i = C_i - A_i$. Because all tasks arrive at $t = 0$, arrival time $A_i = 0$ for every task, meaning turnaround time equals completion time for each task ($T_i = C_i$).

1. FCFS Turnaround Times: $T_A = 6s$, $T_B = 14s$, $T_C = 18s$, $T_D = 28s$, $T_E = 33s$.
   Total Turnaround Time = 6 + 14 + 18 + 28 + 33 = 99 seconds.
   Average Turnaround Time = 99 / 5 = 19.8 seconds.

2. SJF Turnaround Times: $T_A = 15s$, $T_B = 23s$, $T_C = 4s$, $T_D = 33s$, $T_E = 9s$.
   Total Turnaround Time = 4 + 9 + 15 + 23 + 33 = 84 seconds.
   Average Turnaround Time = 84 / 5 = 16.8 seconds.

3. LJF Turnaround Times: $T_A = 24s$, $T_B = 18s$, $T_C = 33s$, $T_D = 10s$, $T_E = 29s$.
   Total Turnaround Time = 10 + 18 + 24 + 29 + 33 = 114 seconds.
   Average Turnaround Time = 114 / 5 = 22.8 seconds.

4. Round Robin (q = 3) Turnaround Times: $T_A = 18s$, $T_B = 29s$, $T_C = 22s$, $T_D = 33s$, $T_E = 27s$.
   Total Turnaround Time = 18 + 29 + 22 + 33 + 27 = 129 seconds.
   Average Turnaround Time = 129 / 5 = 25.8 seconds.

Scheduling Policy	Task Turnaround Times (A, B, C, D, E)	Sum of Turnaround Times	Average Turnaround Time	Performance Rank
Shortest Job First (SJF)	A: 15s, B: 23s, C: 4s, D: 33s, E: 9s	84 seconds	16.8 seconds	Rank 1 (Best Turnaround)
First-Come, First-Served (FCFS)	A: 6s, B: 14s, C: 18s, D: 28s, E: 33s	99 seconds	19.8 seconds	Rank 2
Longest Job First (LJF)	A: 24s, B: 18s, C: 33s, D: 10s, E: 29s	114 seconds	22.8 seconds	Rank 3
Round Robin (RR, q = 3)	A: 18s, B: 29s, C: 22s, D: 33s, E: 27s	129 seconds	25.8 seconds	Rank 4 (Worst Turnaround)

Evaluation: Is Turnaround Time a Good Measure for Scheduling Policies?
Average turnaround time is an effective quantitative benchmark for batch computing systems because it evaluates overall throughput and job latency. Minimizing average turnaround time shortens the duration tasks hold memory, locks, and network sockets, reducing system congestion.

However, turnaround time is not a complete metric on its own for two key reasons:
1. Fairness and Starvation: SJF achieves the best average turnaround time by favoring short jobs, but

in open systems where short jobs arrive continuously, long tasks (like Task D) suffer indefinite starvation.
2. Interactive Responsiveness: In cloud microservices and user facing applications, response time (the delay until a task first gets CPU time) matters more than total completion time. Round Robin grants every task CPU execution within the first 12 seconds, whereas FCFS and SJF force later tasks to sit idle for up to 23 or 28 seconds before receiving a single CPU cycle. In addition, Round Robin incurs context switching overhead that theoretical turnaround calculations ignore.

Evaluation: Which Scheduling Policy Performs the Best?
The best scheduling policy depends directly on the system objective:
• For Minimum Latency and Batch Throughput: Shortest Job First (SJF) performs the best, achieving the lowest average turnaround time of 16.8 seconds. This represents a 15.2% improvement over FCFS (19.8s) and a 34.9% improvement over Round Robin (25.8s). SJF is provably optimal for minimizing average turnaround time when all jobs arrive simultaneously.
• For Fairness and Interactive Systems: Round Robin (RR) performs the best in practical multi tenant environments. Although its average turnaround time is the highest (25.8s) due to time slicing delays across all tasks, it prevents starvation and guarantees bounded response times for every client.
• Longest Job First (LJF) performs the worst (22.8s), penalizing short jobs without providing fairness or responsiveness benefits.